

CSE 4125: Distributed Database Systems

Chapter – 6

(Part – D)

Optimization of Access Strategies.

Semi-join Programs

(Join using Semi-join)

Semi-join Programs

$$R \text{ JN}_{C=A} S \leftrightarrow (R \text{ SJ}_{C=A} \text{ PJ}_A S) \text{ JN}_{C=A} S$$

A	B
1	4
2	5
3	6
3	7

*Assume, $C_0 = 0$, $C_1 = 1$, $\rho = 0.2$
 $\text{Size}(A) = \text{Size}(B) = \text{Size}(C) = \text{Size}(D) = 2 \text{ bytes}$*

C	D
1	6
2	7
4	8
5	9



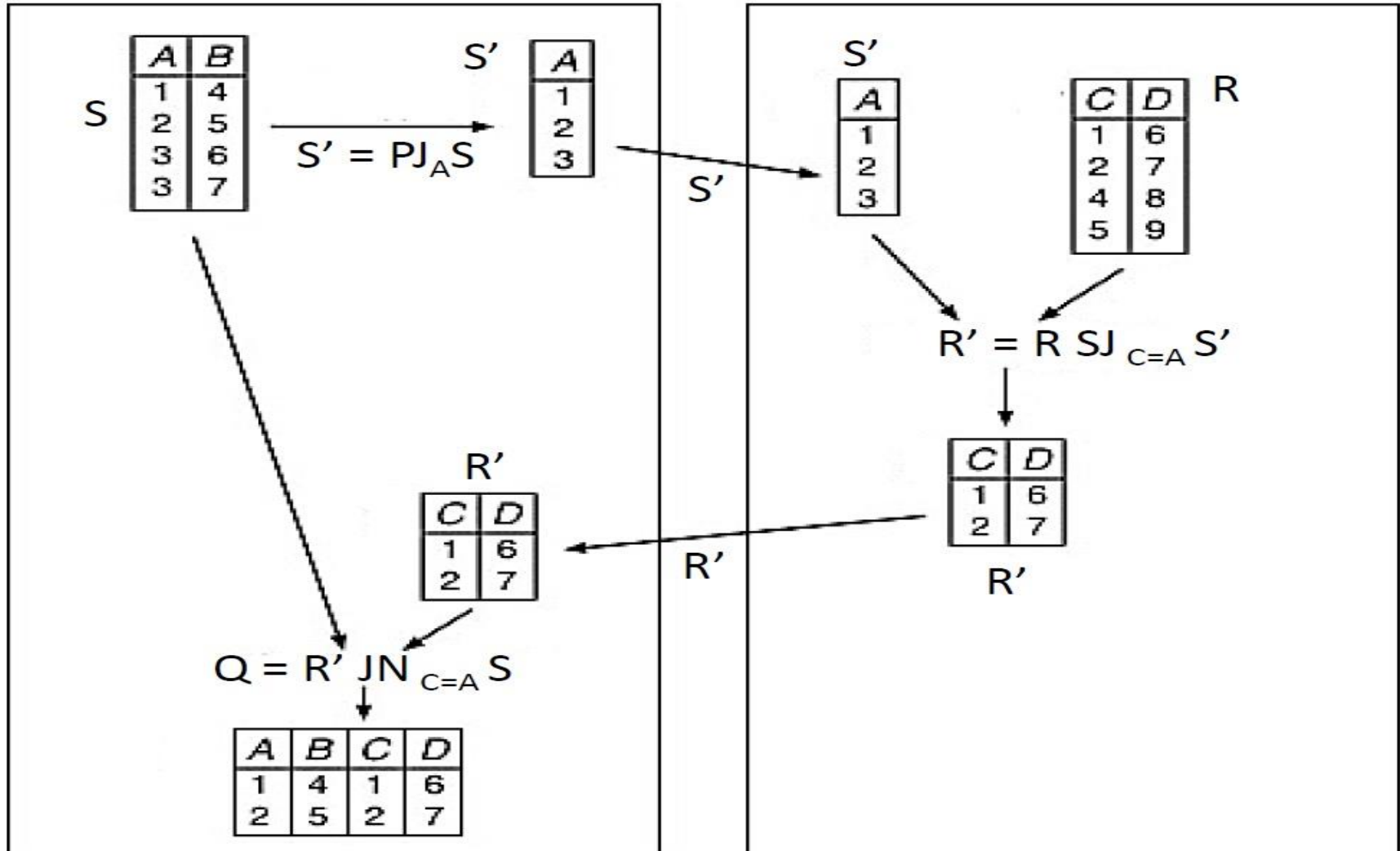
We want to perform $R \text{ JN}_{C=A} S$ at site – 1 using Semi-Join Program.

Semi-join Programs (cont.)

$$R \text{ JN}_{C=A} S \leftrightarrow (R \text{ SJ}_{C=A} \text{ PJ}_A S) \text{ JN}_{C=A} S$$

site 1

site 2

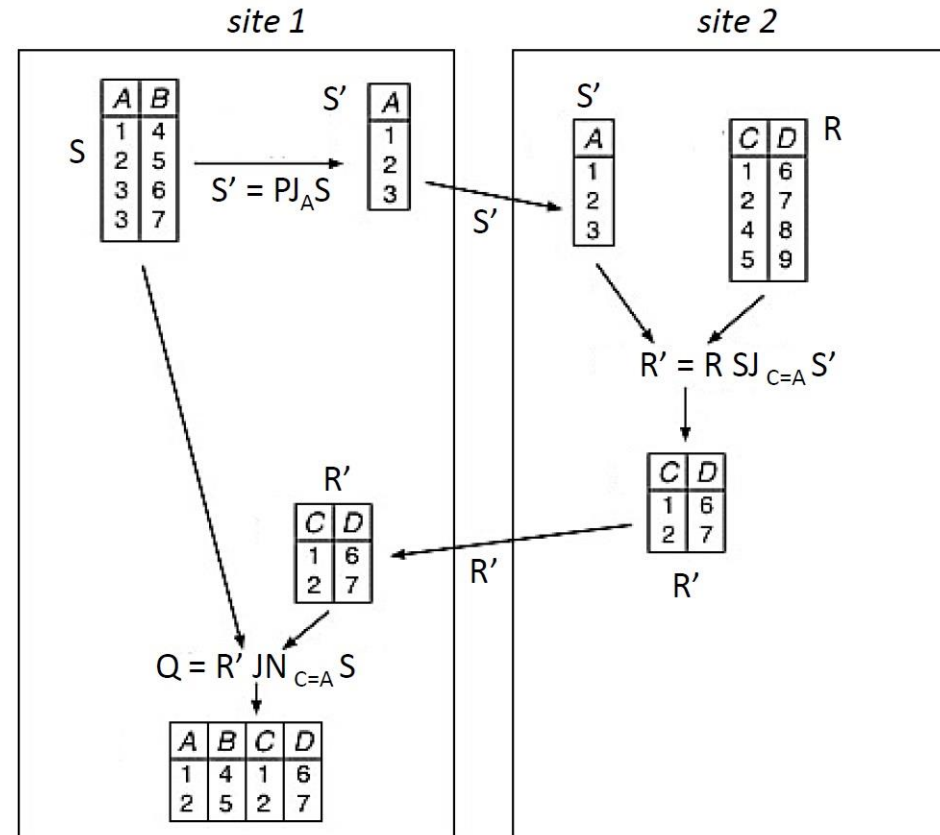


Semi-join Programs (cont.)

Steps of Semi-join program

1. Send $S' = PJ_A(S)$ to **site-2**
2. Compute $R' = R SJ_{C=A} S'$ at **site-2**
3. Send R' to **site-1**
4. Compute $Q = R' JN_{C=A} S$ at **site-1**

Task: What will be the steps if we want to perform the join at site 2?

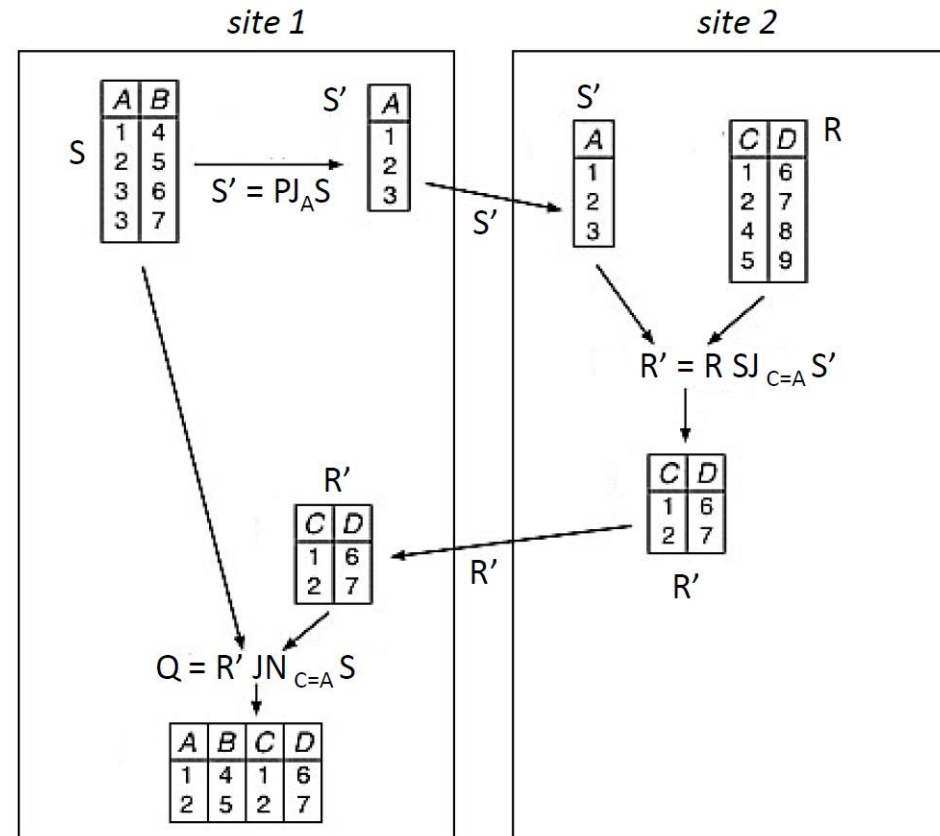


Semi-join Programs (cont.)

Cost of Semi-join program

Step 1: Send $S' = PJ_A(S)$ to **site-2**

$$\begin{aligned}
 TC_1 &= C_0 + C_1 * x \\
 &= C_0 + C_1 * Card(S') * size(S') \\
 &= C_0 + C_1 * val(A[S]) * size(A) \\
 &= 0 + 1 * 3 * 2 \text{ bytes} \\
 &= 6 \text{ bytes} \\
 &= 6 * 8 \text{ bits} \\
 &= 48 \text{ bits}
 \end{aligned}$$

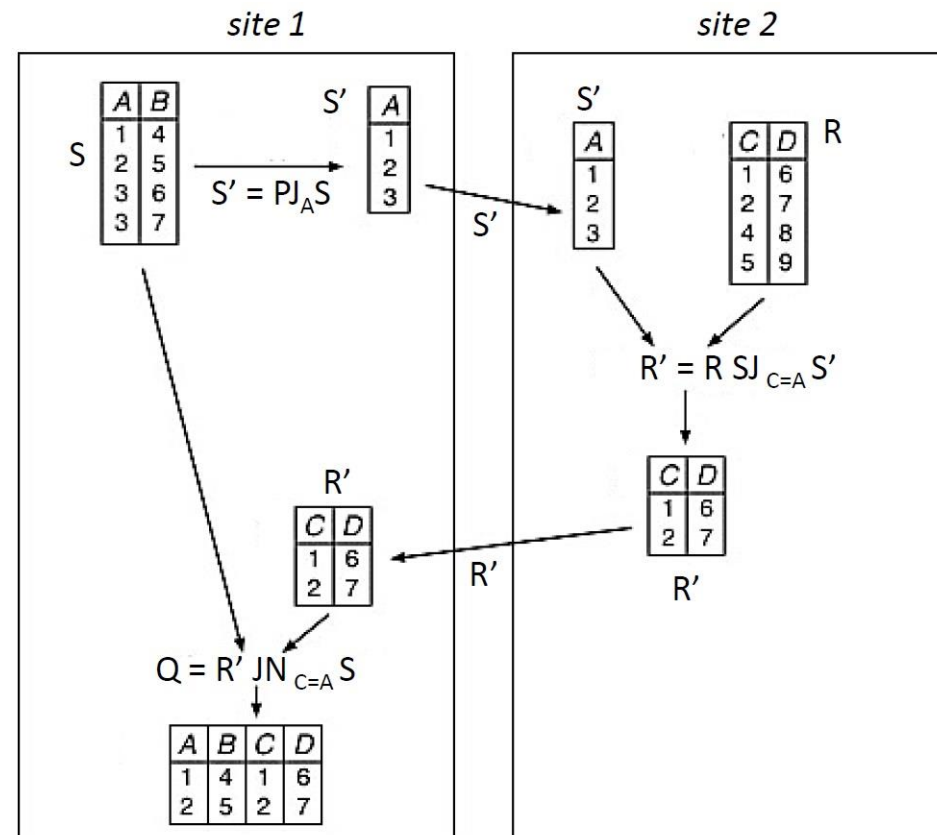


Semi-join Programs (cont.)

Cost of Semi-join program

Step 2: Compute $R' = R \Join_{C=A} S'$ at **site-2**

$$TC_2 = 0$$

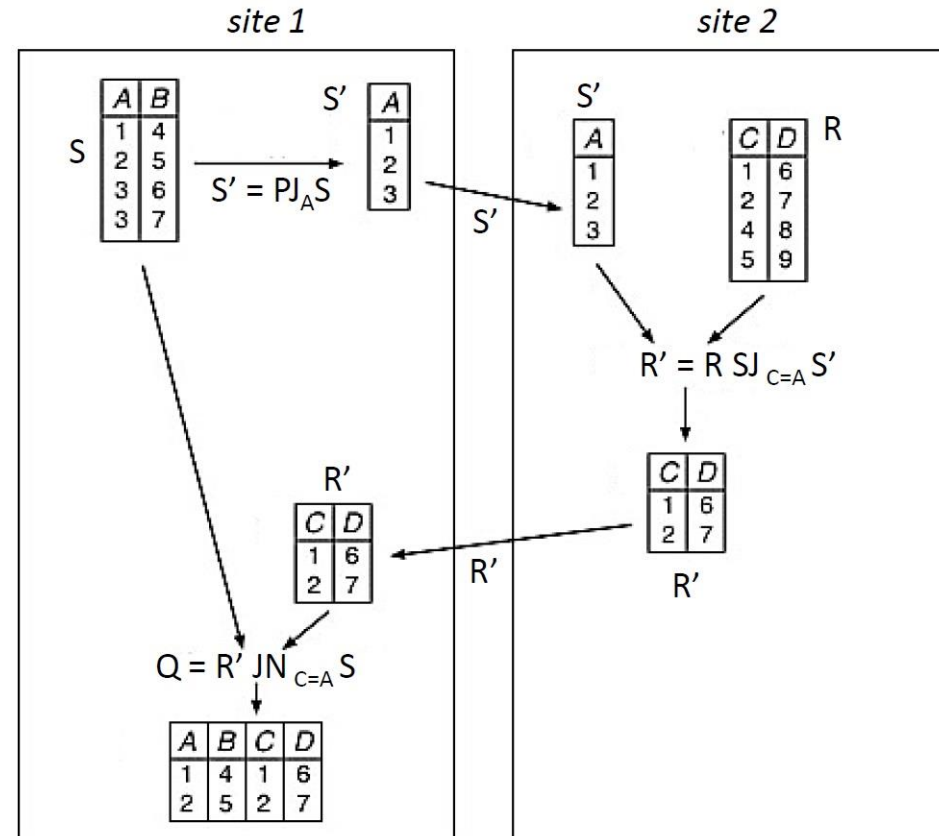


Semi-join Programs (cont.)

Cost of Semi-join program

Step 3: Send R' to **site-1**

$$\begin{aligned}
 TC_3 &= C_0 + C_1 * x \\
 &= C_0 + C_1 * Card(R') * size(R') \\
 &= C_0 + C_1 * \rho * Card(R) * size(R) \\
 &= 0 + 1 * 0.2 * 4 * 4 \text{ bytes} \\
 &= 3.2 \text{ bytes} \\
 &= 3.2 * 8 \text{ bits} \\
 &= 25.6 \text{ bits} \\
 &\approx 26 \text{ bits}
 \end{aligned}$$

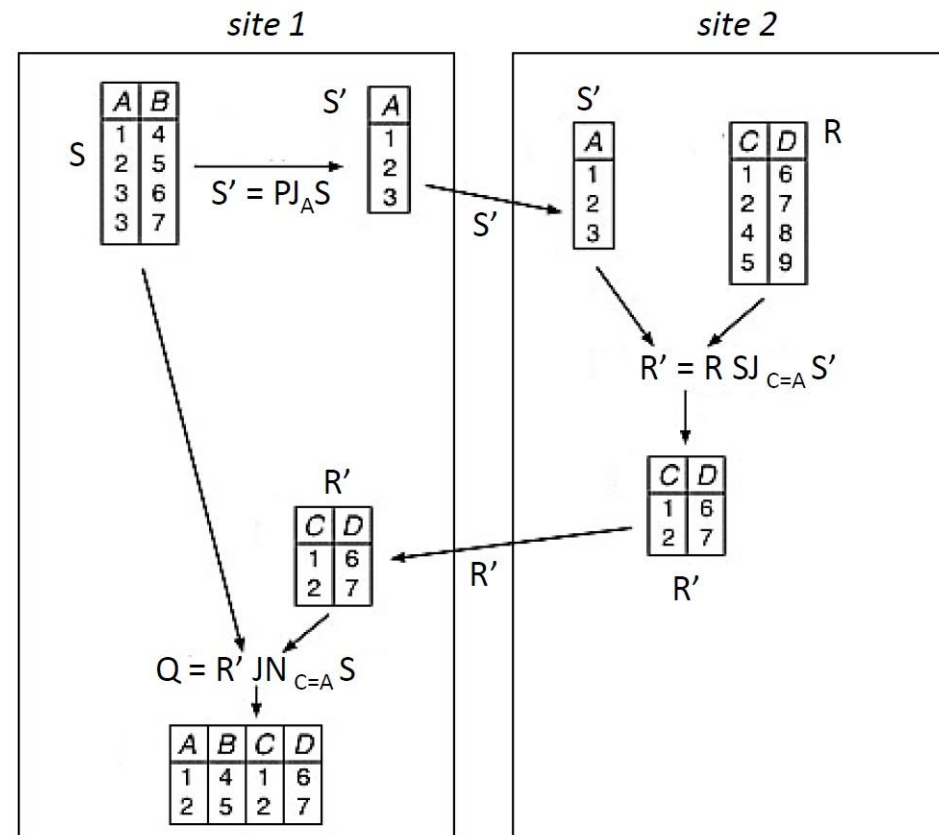


Semi-join Programs (cont.)

Cost of Semi-join program

Step 4: Compute $Q = R' \Join_{C=A} S$ at **site-1**

$$TC_4 = 0$$



Semi-join Programs (cont.)

Cost of Semi-join program

Total cost

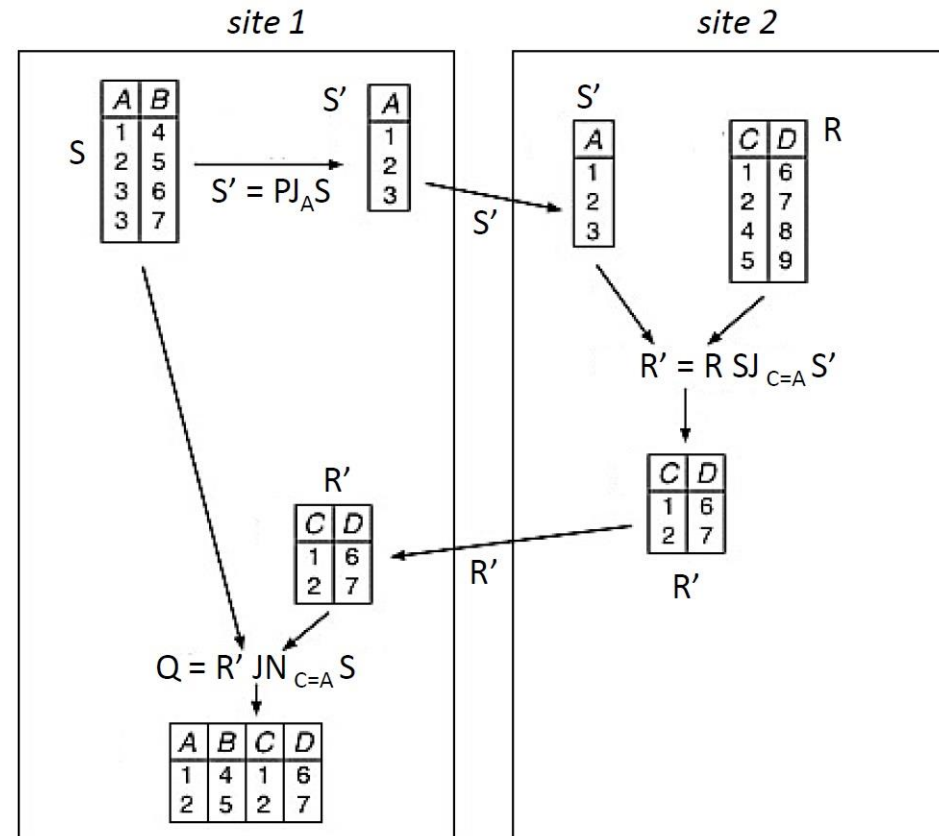
$$TC_{SJ} = TC_1 + TC_2 + TC_3 + TC_4 = 48 + 0 + 26 + 0 = 74 \text{ bits}$$

If $TC_{SJ} < TC_{JN}$ then semi-join program is profitable.

Semi-join Programs (cont.)

Cost **without** Semi-join program

$$\begin{aligned}
 TC_{JN} &= C_0 + C_1 * x \\
 &= C_0 + C_1 * Card(R) * size(R) \\
 &= 0 + 1 * 4 * 4 \text{ bytes} \\
 &= 16 \text{ bytes} \\
 &= 16 * 8 \text{ bits} \\
 &= 128 \text{ bits}
 \end{aligned}$$

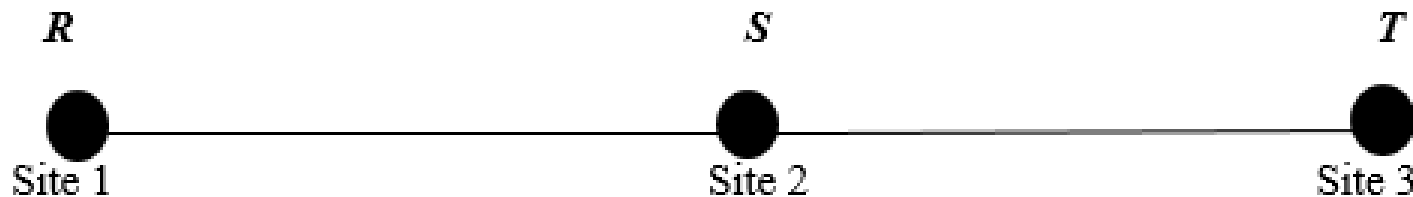


Other Applications of Semi-join Programs

- ❑ Semi-join programs can be used as **fragment reducers** (operations that can reduce cardinality of a relation).
 - Similarly to unary operations.
- ❑ Full reducer:
 - Chain of semi-joins.

Exercise

Consider the following distributed database with relations **R**, **S** and **T** over a network of site 1, 2 and 3.



Assume the following specifications are given.

$$C_0^{12} = C_0^{23} = C_1^{21} = C_1^{32} = 0 \text{ unit}$$

$$C_1^{12} = C_1^{23} = C_0^{21} = C_0^{32} = 1 \text{ unit}$$

size (R) = 20 bytes, **size(T)** = 20 bytes, **size(S)** = 40 bytes, **size(a)** = **size(b)** = 1 byte

card(R) = 100, **card(S)** = 50, **card(T)** = 50

val(a[R]) = val(b[S]) = val(a[T]) = 50

$R \Join_{a=b} S$ has selectivity $\rho = 0.1$

$S SJ_{b=a}R$ has selectivity $\rho = 0.9$

$T SJ_{a=b}S$ has selectivity $\rho = 0.5$

$S SJ_{b=a}T$ has selectivity $\rho = 0.5$

Determine the total transmission cost of performing $(R JN_{a=b} S) DF (T JN_{a=b} S)$ at site 2 using semi-join program only. [C^{xy} means transimission cost from site x to site y]